

Natural Policy Gradient

DIRECTION OF STEEPEST ASCENT/DESCENT

Consider the Euclidean Space

$\frac{\partial f(x,y)}{\partial x}$ and $\frac{\partial f(x,y)}{\partial y}$ are the partial derivatives of $f(x,y)$.

Let vector $\vec{u}$ be a direction vector.

$$\vec{u} = u_x \hat{i} + u_y \hat{j}, \quad u_x^2 + u_y^2 = 1.$$

Suppose we move by Δs along $\vec{u}$ starting at point P .

$$\begin{aligned} \Delta f &= f(P + \Delta s \vec{u}) - f(P) \\ &= f(x + \Delta s u_x, y + \Delta s u_y) - f(x, y) \\ &\approx f(x, y) + \Delta s u_x \frac{\partial f(x, y)}{\partial x} + \Delta s u_y \frac{\partial f(x, y)}{\partial y} \end{aligned}$$



Rate of change

$$\frac{\Delta f}{\Delta s} = u_x \frac{\partial f(x, y)}{\partial x} + u_y \frac{\partial f(x, y)}{\partial y}$$

In the limit as $\Delta s \rightarrow 0$, we get the directional

$$\begin{aligned} \text{derivative } D_{\vec{u}} f &= u_x \frac{\partial f(x, y)}{\partial x} + u_y \frac{\partial f(x, y)}{\partial y} \\ &= [u_x \quad u_y] \begin{bmatrix} \frac{\partial f(x, y)}{\partial x} \\ \frac{\partial f(x, y)}{\partial y} \end{bmatrix}. \end{aligned}$$

For steepest/fastest ascent maximize above, given $u_x^2 + u_y^2 = 1$.

What's the maximizing direction?

What's the vector of interest called?

Geometry Of Statistical Models

*DIFFERENTIAL-GEOMETRIC METHODS IN STATISTICS, Shun-Chi Amari,
Lecture Notes 1985.*

- Think looking for an optimal distribution
 - As in policy gradient or via policy improvement
- We want to find this distribution from the set of all possible distributions
- We restrict to searching among probability distributions that belong to a parametrized statistical model

Geometry Of Statistical Models

- A statistical model embeds a finite-dimensional manifold in the set of all distributions
- Points in the manifold correspond to distributions
- A Gaussian/ Normal statistical model is an example of a parametrized model

Manifold: A space that locally looks Euclidean but has a different structure globally. For example, a sphere, a circle.

Geometry Of Statistical Models

- A statistical model embeds a finite-dimensional manifold in the set of all distributions
- Points in the manifold correspond to distributions
- Distance between distributions is distance between the corresponding points on the manifold

Manifold of a Statistical Model

- Statistical model

$$S = \{p(x, \theta)\}$$

- p is the PDF

- Parameter vector is

$$\theta = (\theta^1, \theta^2, \dots, \theta^n)$$

Manifold of a Statistical Model

- Example parameter vector for the Gaussian RV?

$$(\mu, \sigma)$$

PDF is

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Manifold of a Statistical Model

- For a Gaussian, the set of parameters is the half-plane

$$\{(\mu, \sigma) \mid -\infty < \mu < \infty, 0 < \sigma\}$$

MULTINOMIAL MODEL

A multinomial model is a set of distributions

$$P(x, \theta) = \sum_{i=1}^n \delta(x-i) \theta_i + \delta(x=n+1) (1 - \sum_{i=1}^n \theta_i)$$

where $\theta_1 = p_1, \theta_2 = p_2, \dots, \theta_n = p_n$ [n+1 faced biased coin]

If $n=1$, we have two points in the sample space.

Sample space $S_X = \{0, 1\}$

$\theta_1 = p_1, 0 \leq p_1 \leq 1$.

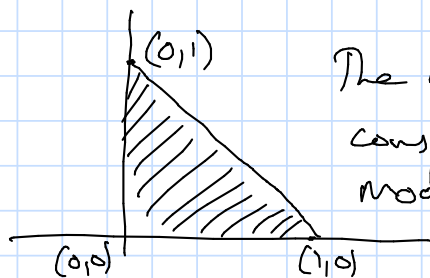


Points p_1 in the interval $[0, 1]$ are distributions over

$S_X = \{0, 1\}$.

Sample space $S = \{0, 1, 2\}$

$n=2$.



The distributions that constitute the statistical model are points in the triangular region.

Compact form of the PMF:

$$P[X=x] =$$

$x \in \{1, 2, \dots, n+1\}$

n-dimensional Manifold

- A *Hausdorff* space that is locally *homeomorphic* to an n-dimensional Euclidean space
- *Hausdorff* space
 - For distinct points in space, there exist neighborhoods that are disjoint

n-dimensional Manifold

- A *Hausdorff* space that is locally *homeomorphic* to an n-dimensional Euclidean space
- U is an open subset of manifold S
- *Locally homeomorphic*
 - If set U is homeomorphic to an n-dim Euclidean space with homeomorphism ϕ
For every $p \in U$, $\phi(p)$ assigns a point $\theta \in R^n$
 - ϕ is one-to-one and onto by the definition of homeomorphism
Every point in U is mapped to R^n & vice-versa.

n-dimensional Manifold

ϕ introduces a coordinate system in U

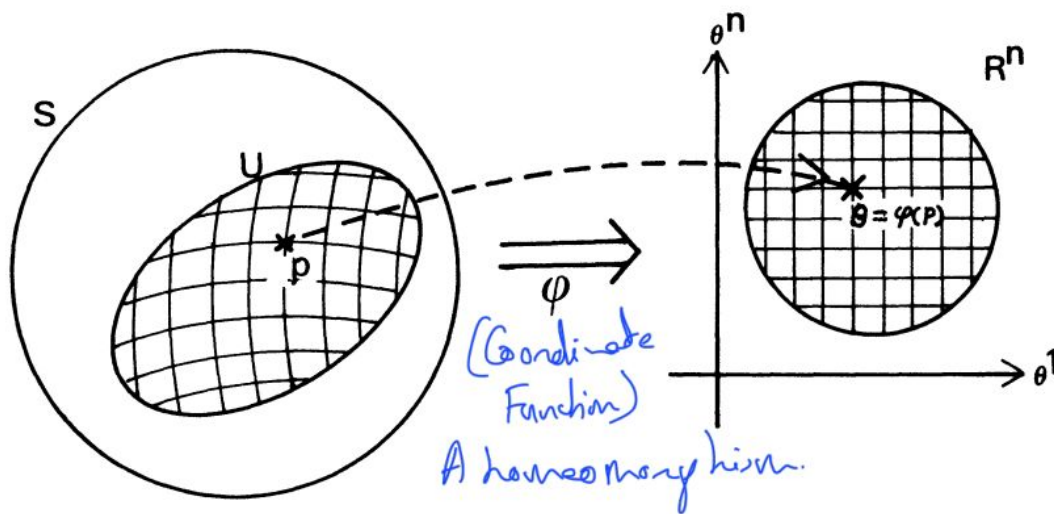
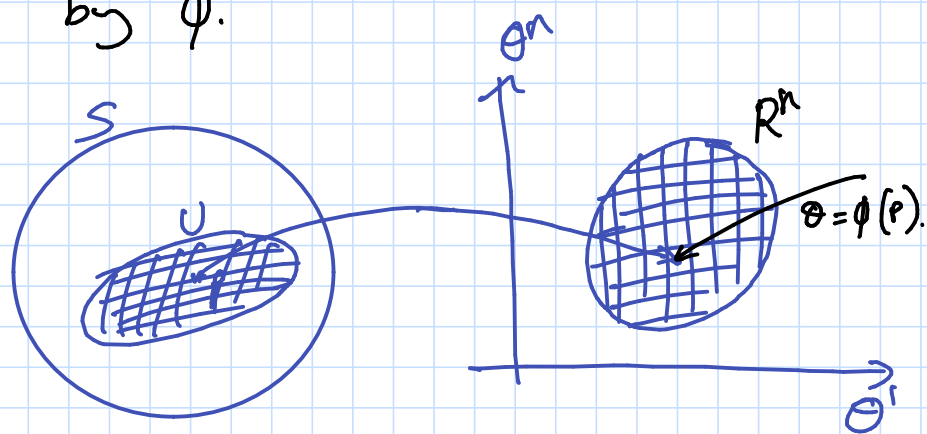


Fig 2.1 in Differential-Geometrical
Methods in Statistics by Shun-ichi Amari

The mapping ϕ is called a co-ordinate function in the co-ordinate neighborhood U . ϕ enables us to introduce a coordinate system in U . A point p in U is given coordinates $\theta = (\theta_1, \theta_2, \dots, \theta_n)$ or shortly $(\theta_i), i=1, \dots, n$.

Coordinate curves in U can be drawn by mapping coordinate curves in $\mathbb{R}^n$ by ϕ^{-1} .



⑧ A local differentiable structure is introduced in U by defining a coordinate system. We will be interested in $\frac{\partial}{\partial \theta_i}, i=1, 2, \dots, n$.

Stereographic projection

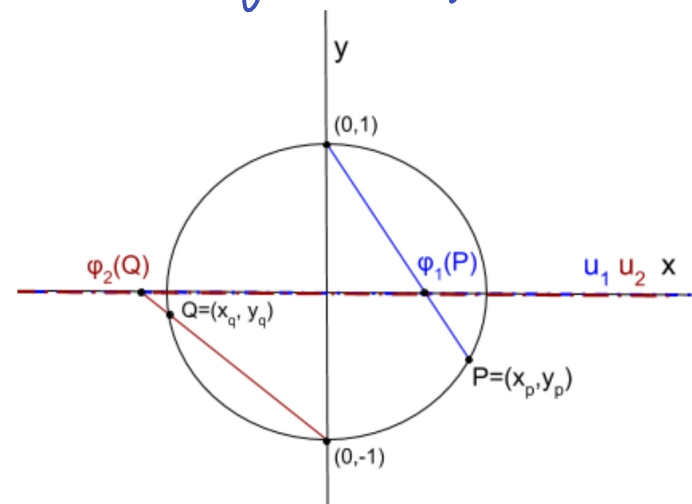


Fig 5 in <https://bjlkeng.io/posts/manifolds/>

TANGENT SPACE (AT point p in S)

Tangent Space: A vector space obtained by local linearization of S at p . Consists of tangent vectors to curves through p .

⊗ We need to understand/define:

- Curves in the manifold S .
- Tangent to curves at a point $p \in S$.
- A basis of the tangent that spans all tangents to curves through a point.

TANGENT SPACE (AT point p in S)

A curve c is a continuous mapping from a closed interval $[a, b] \in \mathbb{R}$ into S .

$$c = c(t) \in S, \quad t \in [a, b].$$

If we use a coordinate system θ , we can define the corresponding curve as $\theta = \theta(t)$.

At t , $\theta(t) = \{\theta^1(t), \theta^2(t), \dots, \theta^n(t)\}$ gives the coordinates of the point $c(t)$.

TANGENT SPACE (AT point p in S)

Now consider a function f over the curve c .

$$f \circ c : [a, b] \rightarrow \mathbb{R}.$$

Let Cf be the derivative of this function along the curve.

$$\begin{aligned} Cf &= \frac{d f \circ c}{dt} = \frac{d f(\theta(t))}{dt} \\ &= \sum_{i=1}^n \frac{d\theta^i}{dt} \frac{\partial f}{\partial \theta^i} \end{aligned}$$

Cf is the directional derivative associated with the curve c .

↳ The rate of change of the function in direction of the tangent to the curve c at $\theta(t)$.

TANGENT SPACE (AT point p in S)

One can think of the operator C in

$$Cf = \sum_{i=1}^n \frac{d\theta^i}{dt} \frac{\partial f(\theta(t))}{\partial \theta^i}$$

as the sum of vectors $\frac{\partial}{\partial \theta^i}$ scaled

by their components $\frac{d\theta^i}{dt}$.

$\frac{\partial}{\partial \theta^i}$ is the rate of change w.r.t the

i th coordinate of the curve $\theta(t)$.

We want to:

- Create a basis at point $p \in S$
that spans the tangent space T_p at p .
- Tangent space is the set of all
tangent vectors of curves through p .

Construct n curves through point $p_0 \in S$.
(indexed $(i), i=1, \dots, n$)

In the coordinate system we
created using the homeomorphism
 ϕ ,
 p_0 is $\theta_0 = (\theta_0^1, \theta_0^2, \dots, \theta_0^n)$.

$$\text{Curve } \theta_{(1)}(t) = (\theta_0^1 + t, \theta_0^2, \dots, \theta_0^n)$$

$$\frac{d}{dt} f(\theta_{(1)}(t)) = \frac{\partial f}{\partial \theta^1}$$

$$\vdots$$
$$\text{Curve } \theta_{(i)}(t) = (\theta_0^1, \theta_0^2, \dots, \theta_0^i + t, \dots, \theta_0^n)$$

$$\frac{d}{dt} f(\theta_{(i)}(t)) = \frac{\partial f}{\partial \theta^i}, \quad i=1, 2, \dots, n.$$

FACT:

$\partial_i \triangleq \frac{\partial}{\partial x^i}$, $i=1, 2, \dots, n$ constitute a basis.

Any directional derivative operator A can be written as:

$$\left. \begin{aligned} A &= \sum_{i=1}^n A^i \frac{\partial}{\partial x^i} \\ &= \sum_{i=1}^n A^i \partial_i \end{aligned} \right\} \begin{array}{l} A^i \text{ is the} \\ \text{component of} \\ A \text{ along} \\ \frac{\partial}{\partial x^i} \end{array}$$

We have the vector space

$$T_p = \left\{ A : A = \sum_{i=1}^n A^i \partial_i \right\}$$

Ⓡ Every linear combination of ∂_i is a directional derivative at $p \in M$. Recall $\partial(p) = 0$.

Ⓡ Think of T_p above as T_p dot product. We can write:

$$\nabla_p^T \begin{bmatrix} A_i \end{bmatrix}$$